

LOS_Lab Assignment 2

Srinivas P Bodke
261100690018

The main aim of this lab session is to provide hands-on experience on:

1. Getting Help

Task	Command Name	Syntax	Example
To get manual page for the known command	man	man <command>	man ls
<pre>LS(1) User Commands LS(1) NAME ls - list directory contents SYNOPSIS ls [OPTION]... [FILE]... DESCRIPTION List information about the FILES (the current directory by default). Sort entries alphabetically if none of -cftuvSUX nor --sort is specified. Mandatory arguments to long options are mandatory for short options too. -a, --all do not ignore entries starting with . -A, --almost-all do not list implied . and .. --author with -l, print the author of each file -b, --escape print C-style escapes for nongraphic characters --block-size=SIZE with -l, scale sizes by SIZE when printing them; e.g., '--block-size=M'; see SIZE format below Manual page ls(1) line 1 (press h for help or q to quit)</pre>			
To get manual page for the unknown command	apropos	apropos	apropos copy

```

openssl-s_time (1ssl) - SSL/TLS performance timing program
openssl-sess_id (1ssl) - SSL/TLS session handling command
pam_exec (8) - PAM module which calls an external command
pam_shells (8) - PAM module to check for valid login shell
pkaction (1) - Get details about a registered action
proc_dma (5) - ISA DMA channels
proc_iide (5) - IDE channels and attached devices
proc_ksyms (5) - kernel exported symbols
protocols (5) - protocols definition file
purge-old-kernels (1) - remove old kernel and header packages from the system
remove-shell (8) - remove shells from the list of valid login shells
rtacct (8) - network statistics tools.
securetty (5) - list of terminals on which root is allowed to login
shells (5) - pathnames of valid login shells
signal (7) - overview of signals
ssl (7ssl) - OpenSSL SSL/TLS library
strip (1) - discard symbols and other data from object files
systemd-cgls (1) - Recursively show control group contents
systemd-creds (1) - Lists, shows, encrypts and decrypts service credentials
systemd.system-credentials (7) - System Credentials
tc-cbq-details (8) - Class Based Queueing
textdomain (3) - set domain for future gettext() calls
update-shells (8) - update the list of valid login shells
watch (1) - execute a program periodically, showing output fullscreen
which.debianutils (1) - locate a command
x86_64-linux-gnu-c++filt (1) - demangle C++ and Java symbols
x86_64-linux-gnu-nm (1) - list symbols from object files
x86_64-linux-gnu-strip (1) - discard symbols and other data from object files
zipdetails (1) - display the internal structure of zip files
root@DESKTOP-E4HFS1K:~#

```

To know the source file binary	whereis	whereis <command>	whereis ls
<pre> root@DESKTOP-E4HFS1K:~# whereis ls ls: /usr/bin/ls /usr/share/man/man1/ls.1.gz root@DESKTOP-E4HFS1K:~# </pre>			
To know the path of the command	which	which <command>	which cd
<pre> srinivas@DESKTOP-E4HFS1K:/root\$ which cd cd: shell built-in command </pre>			
To know the command is external or internal	type	type<command>	type cd
<pre> srinivas@DESKTOP-E4HFS1K:/root\$ type cd cd is a shell builtin srinivas@DESKTOP-E4HFS1K:/root\$ </pre>			
To get help for the internal command	help	help <command>	help cd

```

srinivas@DESKTOP-E4HFS1K:/root$ help cd
cd: cd [-L|[-P [-e]] [-@]] [dir]
    Change the shell working directory.

    Change the current directory to DIR. The default DIR is the value of the
    HOME shell variable. If DIR is "-", it is converted to $OLDPWD.

    The variable CDPATH defines the search path for the directory containing
    DIR. Alternative directory names in CDPATH are separated by a colon (:).
    A null directory name is the same as the current directory. If DIR begins
    with a slash (/), then CDPATH is not used.

    If the directory is not found, and the shell option `cdable_vars' is set,
    the word is assumed to be a variable name. If that variable has a value,
    its value is used for DIR.

Options:
  -L      force symbolic links to be followed: resolve symbolic
          links in DIR after processing instances of `..'
  -P      use the physical directory structure without following
          symbolic links: resolve symbolic links in DIR before
          processing instances of `..'
  -e      if the -P option is supplied, and the current working
          directory cannot be determined successfully, exit with
          a non-zero status
  -@      on systems that support it, present a file with extended
          attributes as a directory containing the file attributes

The default is to follow symbolic links, as if `-L' were specified.
`..' is processed by removing the immediately previous pathname component
back to a slash or the beginning of DIR.

Exit Status:
Returns 0 if the directory is changed, and if $PWD is set successfully when
-P is used; non-zero otherwise.
srinivas@DESKTOP-E4HFS1K:/root$

```

**To list out bash
commands**

compgen

compgen - c

compgen -c

```

x86_64-linux-gnu-objdump
x86_64-linux-gnu-ranlib
x86_64-linux-gnu-readelf
x86_64-linux-gnu-size
x86_64-linux-gnu-strings
x86_64-linux-gnu-strip
xargs
xauth
xdg-user-dir
xdg-user-dirs-update
xsubpp
xxd
xz
xzcat
xzcmp
xzdiff
xzegrep
xzfgrep
xzgrep
xzless
xzmore
yes
ypdomainname
zcat
zcmp
zdiff
zdump
zegrep
zfgrep
zforce
zgrep
zipdetails
zless
zmore
znew
[
srinivas@DESKTOP-E4HFS1K:/root$

```

**To know the usage of
the command**

whatis

whatis <command>

whatis ls

```

srinivas@DESKTOP-E4HFS1K:/root$ whatis ls
ls (1)          - list directory contents
srinivas@DESKTOP-E4HFS1K:/root$

```

2. Basic commands

Task	Command Name	Syntax	Example
To know today's date	date	date	date
<pre>srinivas@DESKTOP-E4HFS1K:/root\$ date Sun Aug 23 13:21:44 IST 2026 srinivas@DESKTOP-E4HFS1K:/root\$ _</pre>			
To print calendar	cal	cal [month] [year]	cal
<pre>(srinivas@DESKTOP-E4HFS1K: [~]) \$-cal August 2026 Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31</pre>			
To print kernel version	uname	uname -r	uname -r
<pre>srinivas@DESKTOP-E4HFS1K:/root\$ uname -r 4.4.0-19041-Microsoft srinivas@DESKTOP-E4HFS1K:/root\$ _</pre>			
To print default shell	echo	echo \$SHELL	echo \$SHELL
<pre>srinivas@DESKTOP-E4HFS1K:/root\$ echo \$SHELL /bin/bash srinivas@DESKTOP-E4HFS1K:/root\$</pre>			
To print currently logged in user	whoami	whoami	whoami
<pre>srinivas@DESKTOP-E4HFS1K:/root\$ whoami srinivas srinivas@DESKTOP-E4HFS1K:/root\$</pre>			

To create shortcut for command	alias	alias name='command'	alias l='ls -l'
<pre> root@DESKTOP-E4HFS1K:~# alias l='ls -l' root@DESKTOP-E4HFS1K:~# l total 0 drwxr-xr-x 1 root root 4096 Aug 22 14:50 LOS_ASS1 drwxr-xr-x 1 root root 4096 Aug 13 20:33 demo drwxr-xr-x 1 root root 4096 Aug 13 20:12 linux_demo drwxr-xr-x 1 root root 4096 Aug 13 20:24 linux_test -r---w---x 1 root root 30 Aug 22 17:24 sample.txt dr----- 1 root root 4096 Aug 22 17:18 sample_dir root@DESKTOP-E4HFS1K:~# </pre>			
To delete shortcut	unalias	unalias name	unalias l
<pre> root@DESKTOP-E4HFS1K:~# unalias l root@DESKTOP-E4HFS1K:~# l l: command not found root@DESKTOP-E4HFS1K:~# _ </pre>			
To change the timestamp of the file	touch	touch <file>	touch -m time2.txt
<pre> root@DESKTOP-E4HFS1K:~# touch -m time2.txt root@DESKTOP-E4HFS1K:~# ls LOS_ASS1 demo linux_demo linux_test sample.txt sample_dir time2.txt root@DESKTOP-E4HFS1K:~# _ </pre>			
To create empty files	touch	touch file1 file2	touch lab2.txt lab2.c
<pre> root@DESKTOP-E4HFS1K:~# touch lab2.txt lab2.c root@DESKTOP-E4HFS1K:~# ls LOS_ASS1 demo lab2.c lab2.txt linux_demo linux_test sample.txt sample_dir time2.txt root@DESKTOP-E4HFS1K:~# _ </pre>			
To clear the screen	clear	clear	clear

<pre>root@DESKTOP-E4HFS1K:~#</pre>			
To know disk usage	du	du [options] [path]	du -sh .
<pre>root@DESKTOP-E4HFS1K:~# du -sh 16K . root@DESKTOP-E4HFS1K:~#</pre>			
To know free space in the system	df	df [options]	df -h
<pre>root@DESKTOP-E4HFS1K:~# df -h Filesystem Size Used Avail Use% Mounted on rootfs 285G 159G 126G 56% / none 285G 159G 126G 56% /dev none 285G 159G 126G 56% /run none 285G 159G 126G 56% /run/lock none 285G 159G 126G 56% /run/shm none 285G 159G 126G 56% /run/user tmpfs 285G 15G 270G 5% /tmp C:\ 285G 159G 126G 56% /mnt/c D:\ 144G 18G 126G 13% /mnt/d E:\ 49G 46G 2.8G 95% /mnt/e root@DESKTOP-E4HFS1K:~#</pre>			
To know about the linux release	cat /etc/os-release	cat /etc/os-release	cat /etc/os-release
<pre>root@DESKTOP-E4HFS1K:~# cat /etc/os-release PRETTY_NAME="Ubuntu 24.04.4 LTS" NAME="Ubuntu" VERSION_ID="24.04" VERSION="24.04.4 LTS (Noble Numbat)" VERSION_CODENAME=noble ID=ubuntu ID_LIKE=debian HOME_URL="https://www.ubuntu.com/" SUPPORT_URL="https://help.ubuntu.com/" BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/" PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy" UBUNTU_CODENAME=noble LOGO=ubuntu-logo root@DESKTOP-E4HFS1K:~#</pre>			
To show login history	last	last	last

<pre> root@DESKTOP-E4HFS1K:~# last wtmp begins Tue Aug 11 23:42:35 2026 root@DESKTOP-E4HFS1K:~# </pre>			
To check the current user privileges	sudo	sudo -l	sudo -l
<pre> root@DESKTOP-E4HFS1K:~# sudo -l Matching Defaults entries for root on DESKTOP-E4HFS1K: env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\ use_pty User root may run the following commands on DESKTOP-E4HFS1K: (ALL : ALL) ALL root@DESKTOP-E4HFS1K:~# </pre>			
To find about the Linux release	lsb_release	lsb_release -a	lsb_release -a
<pre> root@DESKTOP-E4HFS1K:~# lsb_release -a No LSB modules are available. Distributor ID: Ubuntu Description: Ubuntu 24.04.4 LTS Release: 24.04 Codename: noble root@DESKTOP-E4HFS1K:~# </pre>			
To check system uptime	uptime	uptime	uptime
<pre> root@DESKTOP-E4HFS1K:~# uptime 17:43:00 up 25 min, 0 user, load average: 0.52, 0.58, 0.59 root@DESKTOP-E4HFS1K:~# </pre>			
To shutdown	shutdown	sudo shutdown -h now	sudo shutdown -h now
<pre> root@DESKTOP-E4HFS1K:~# sudo shutdown -h now </pre>			

To restart	reboot	sudo reboot	sudo reboot
<pre>root@DESKTOP-E4HFS1K:~# sudo reboot</pre>			

3. Navigation

Task	Command	Syntax	Example
To navigate home directory	cd	cd ~	cd ~
<pre>root@DESKTOP-E4HFS1K:~/linux_demo# cd ~ root@DESKTOP-E4HFS1K:~# cd ~ root@DESKTOP-E4HFS1K:~#</pre>			
To navigate to the parent directory	cd	cd ..	cd ..
<pre>root@DESKTOP-E4HFS1K:~/linux_demo# cd .. root@DESKTOP-E4HFS1K:~# cd .. root@DESKTOP-E4HFS1K:/#</pre>			
To navigate to the child directory	cd	cd <directory>	cd home
<pre>root@DESKTOP-E4HFS1K:/# cd home/ root@DESKTOP-E4HFS1K:/home#</pre>			
Alternate command to cd	pushd	pushd <directory>	pushd sample_dir
<pre>root@DESKTOP-E4HFS1K:/home/srinivas# pushd sample_dir/ /home/srinivas/sample_dir /home/srinivas root@DESKTOP-E4HFS1K:/home/srinivas/sample_dir#</pre>			

To go back to the previous directory	cd -	cd -	cd -
<pre>root@DESKTOP-E4HFS1K:/home/srinivas/sample_dir# cd - /home/srinivas root@DESKTOP-E4HFS1K:/home/srinivas#</pre>			
To go to the root directory	cd	cd /	cd /
<pre>root@DESKTOP-E4HFS1K:/home/srinivas# cd / root@DESKTOP-E4HFS1K:/#</pre>			

4. File System

Task	Syntax	Command
How to identify the file system	df -T [path]	df -T .
<pre>root@DESKTOP-E4HFS1K:/# df -T . Filesystem Type 1K-blocks Used Available Use% Mounted on rootfs wslfs 298707152 166557392 132149760 56% / root@DESKTOP-E4HFS1K:/#</pre>		

5. Globbing

Create a dedicated practice directory containing sample files such as:

index.html, login.html, admin.htm, report.txt, access.log, error.log, user1.txt, user2.log, file123, backup1.html, backup2.txt

No	Task	Pattern
a	List all files ending in .html.	ls *.html
b	List all files other than .html files using extended globbing.	ls !(*.html)
c	List filenames containing at least one numeric digit.	ls *[0-9]*
d	List files whose names begin with user followed by a single digit.	ls user[0-9].*
e	List files ending in either .txt or .log using a single extended glob pattern.	ls +(*.txt *.log)
f	List files that do not end in .txt or .log.	ls *.{txt,log}
g	List files beginning with either access or error using extended globbing.	ls {access*,error*}
h	List .html files whose filenames contain a numeric digit.	ls *[0-9]*.html

Result:

```
root@DESKTOP-E4HFS1K:/home/srinivas# ls *.html
1.html 2.html 3.html 5.html A.html B.html C.html a.html b.html d.html
root@DESKTOP-E4HFS1K:/home/srinivas# ls !(*.html)
access.log admin.htm error.log file123 report.txt sample.txt user1.log user2.log

sample_dir:
inside.txt
root@DESKTOP-E4HFS1K:/home/srinivas# ls *[0-9]*
1.html 2.html 3.html 5.html file123 user1.log user2.log
root@DESKTOP-E4HFS1K:/home/srinivas# ls user[0-9].*
user1.log user2.log
root@DESKTOP-E4HFS1K:/home/srinivas# ls +(*.txt|*.log)
access.log error.log report.txt sample.txt user1.log user2.log
root@DESKTOP-E4HFS1K:/home/srinivas# ls *.{txt,log}
access.log error.log report.txt sample.txt user1.log user2.log
root@DESKTOP-E4HFS1K:/home/srinivas# ls {access*ls,error*}
ls: cannot access 'access*ls': No such file or directory
error.log
root@DESKTOP-E4HFS1K:/home/srinivas# ls {access*,err*}
access.log error.log
root@DESKTOP-E4HFS1K:/home/srinivas# ls *[0-9]*.html
1.html 2.html 3.html 5.html
root@DESKTOP-E4HFS1K:/home/srinivas#
```

6. Extended Globbing

1. Assume a directory contains:

access.log, error.log, auth.log, auth.log.1, auth.log.2, system.log, notes.txt

No	Task	Pattern
a	Write an extended glob pattern that selects only access.log, error.log, and auth.log.	ls @(access.log error.log auth.log)
b	Select all files except notes.txt.	ls !(@(notes.txt))
c	Select files named either auth.log.1 or auth.log.2 using pattern matching rather than writing both complete filenames separately.	ls auth.log.@(1 2)

Result:

```
root@DESKTOP-E4HFS1K:/home/srinivas# ls auth.log.@(1|2)
auth.log.1 auth.log.2
root@DESKTOP-E4HFS1K:/home/srinivas# ls !(@(notes.txt))
1.html 5.html C.html admin.htm auth.log.2 b.html file123 system.log
2.html A.html a.html auth.log auth.log.3 d.html report.txt user1.log
3.html B.html access.log auth.log.1 auth1.log error.log sample.txt user2.log

sample_dir:
inside.txt
root@DESKTOP-E4HFS1K:/home/srinivas# ls @(access.log|error.log|auth.log)
access.log auth.log error.log
root@DESKTOP-E4HFS1K:/home/srinivas# _
```

2. Given the files:

malware1.log, malware2.log, malwareA.log, malware_report.txt, normal.log

No	Task	Pattern
a	construct a glob pattern that identifies .log files whose names start with malware and contain a numeric character.	ls malware*[0-9]*.log

Result:

```
root@DESKTOP-E4HFS1K:/home/srinivas# ls malware*[0-9]*.log
malware1.log malware2.log
root@DESKTOP-E4HFS1K:/home/srinivas#
```

3. Given:

failed_login.log, successful_login.log, login_backup.log, login.txt, firewall.log

No	Task	Pattern
a	use extended globbing to select .log files containing either failed_login or successful_login.	ls @(failed_login successful_login).log
b	Construct a pattern that lists all .log files except firewall.log.	ls !(@(firewall.log))

Result:

```
root@DESKTOP-E4HFS1K:/home/srinivas# ls @(failed_login|successful_login).log
failed_login.log  successful_login.log
root@DESKTOP-E4HFS1K:/home/srinivas# ls !(@(firewall.log))
1.html  A.html  access.log  auth.log.2  d.html  malware1.log  successful_login.log
2.html  B.html  admin.htm  auth.log.3  error.log  malware2.log  system.log
3.html  C.html  auth.log  auth1.log  failed_login.log  report.txt  user1.log
5.html  a.html  auth.log.1  b.html  file123  sample.txt  user2.log

sample_dir:
inside.txt
root@DESKTOP-E4HFS1K:/home/srinivas# _
```

7. Filters

No	Task	Command
1	Display the first 10 entries of login.log.	head -n 10 login.log
2	Display the first 5 entries of access.log	head -n 5 access.log
3	Display the first 15 security events from security.log.	head -n 15 security.log
4	Display the first 20 events recorded in incident.log.	head -n 20 incident.log

5	Display only the first entry in login.log.	head -n 1 login.log
6	Display the first 25 web requests recorded in access.log.	head -n 25 access.log
7	Display the last 10 entries of login.log.	tail -n 10 login.log
8	Display the last 5 entries of access.log.	tail -n 5 access.log
9	Display the 20 most recent security events from security.log.	tail -n 20 security.log
10	Display the 25 most recent events from incident.log.	tail -n 25 incident.log
11	Display only the last entry in security.log.	tail -n 1 security.log
12	Display the last 50 events recorded in incident.log.	tail -n 50 incident.log
13	Display the contents of login.log starting from line 41 until the end of the file.	tail -n +41 login.log
14	Display the contents of security.log starting from line 101 until the end of the file.	tail -n +101 security.log
15	Find the total number of lines in login.log.	wc -l login.log
16	Find the total number of lines in access.log.	wc -l access.log
17	Find the total number of security events in security.log.	wc -l security.log

18	Determine the total number of events recorded in incident.log.	wc -l incident.log
19	Count the number of words in login.log.	wc -w login.log
20	Count the number of bytes in security.log.	wc -c security.log
21	Display the line, word, and byte counts of access.log.	wc -l -w -c access.log
22	Find the line counts of all four .log files using a single wc command.	wc -l *.log
23	Display the first 20 entries of login.log and, from those entries, display only the last 5.	head -n 20 login.log tail -n 5
24	Display the last 20 entries of login.log and, from those entries, display only the first 5.	tail -n 20 login.log head -n 5
25	Display the first 30 entries of access.log and then display only the last 10 entries from that output	head -n 30 access.log tail -n 10
26	Display the last 30 entries of access.log and then display only the first 10 entries.	tail -n 30 access.log head -n 10
27	Display the first 50 events from security.log and then display the last 10 events from those 50.	head -n 50 security.log tail -n 10
28	Display the first 50 events from security.log and then display the first 10 events from those 50.	tail -n 50 security.log head -n 10
29	Display entries from line 11 to line 20 of login.log.	tail -n +11 login.log head -n 10 or sed -n '11,20p' login.log

30	Display entries from line 21 to line 30 of login.log.	tail -n +21 login.log head -n 10 or sed -n '21,30p' login.log
31	Display entries from line 11 to line 20 of access.log.	tail -n +11 access.log head -n 10 or sed -n '11,20p' access.log
32	Display entries from line 31 to line 40 of access.log.	tail -n +31 access.log head -n 10 or sed -n '31,40p' access.log
33	Display security events from line 21 to line 30 of security.log.	tail -n +21 security.log head -n 10 or sed -n '21,30p' security.log
34	Display security events from line 51 to line 60 of security.log.	tail -n +51 security.log head -n 10 or sed -n '51,60p' security.log
35	Display incident events from line 101 to line 110 of incident.log.	tail -n +101 incident.log head -n 10 or sed -n '101,110p' incident.log
36	Display incident events from line 201 to line 220 of incident.log.	tail -n +201 incident.log head -n 20 or sed -n '201,220p' incident.log
37	Display incident events from line 491 to line 500.	tail -n +491 incident.log head -n 10 or sed -n '491,500p' incident.log

38	Take the first 20 entries of login.log and count how many lines are displayed.	head -n 20 login.log wc -l
39	Take the last 15 entries of access.log and count how many lines are displayed.	tail -n 15 access.log wc -l
40	Take the first 50 entries of security.log and count the number of words in those entries.	head -n 50 security.log wc -w
41	Take the last 25 entries of security.log and count the number of lines.	tail -n 25 security.log wc -l
42	Extract lines 21–30 from login.log and count the number of extracted lines.	tail -n +21 login.log head -n 10 wc -l or sed -n '21,30p' login.log wc -l
43	Extract lines 51–60 from access.log and count the number of extracted lines.	tail -n +51 access.log head -n 10 wc -l or sed -n '51,60p' access.log wc -l
44	Extract events 101–150 from incident.log and count how many events are displayed	tail -n +101 incident.log head -n 50 wc -l or sed -n '101,150p' incident.log wc -l
45	Display the first 100 events from incident.log, take the last 20 events from that output, and count the number of lines.	head -n 100 incident.log tail -n 20 wc -l